

Gini Impurity - Part 1

Step 1: List all possible split points for study hours

Student	Study hours	Pass (1) / Fail (0)
1	2	0
2	4	0
3	6	1
4	8	1
5	10	1

Possible splits are between the study hour values

- Between 2 and 4 : $(2+4)/2 = 3$
- Between 4 and 6 : $(4+6)/2 = 5$
- Between 6 and 8 : $(6+8)/2 = 7$
- Between 8 and 10 : $(8+10)/2 = 9$

Possible split points are 3, 5, 7, 9

Step 2: For each possible split, compute the Gini impurity for the resulting groups ("Study hours $\leq$ split", "Study hours $>$ split").

$$\text{Gini} = 1 - \sum p(i)^2$$

Split at 3:

(Study hours ≤ 3) $\Rightarrow$ pass = 0, fail = 1

Student 1 (2 hours)

$$\text{Gini} = 1 - (1)^2 = 0$$

(Study hours > 3) $\Rightarrow$ 2, 3, 4, 5 (4, 6, 8, 10)
pass = 3, fail = 1

$$\text{Gini} = 1 - (3/4)^2 - (1/4)^2$$

$$= 1 - 0.5625 - 0.0625$$

$$= 0.375$$

Split at 5:

(Study hours ≤ 5) $\Rightarrow$ 1, 2 (2, 4) $\Rightarrow$ fail = 2

$$\text{Gini} = 1 - (1)^2 = 0$$

(Study hours > 5) $\Rightarrow$ 3, 4, 5 (6, 8, 10)
pass = 3

$$\text{Gini} = 1 - (1)^2 = 0$$

Split at 7:

(Study hours ≤ 7) $\Rightarrow$ 1, 2, 3 (2, 4, 6)

pass = 1, fail = 2

$$\text{Gini} = 1 - (1/3)^2 - (2/3)^2 = 1 - 0.111 - 0.444 = 0.445$$

(Study hours > 7) $\Rightarrow$ 4, 5 (8, 10) $\Rightarrow$ pass = 2

$$\text{Gini} = 1 - (1)^2 = 0$$

Split at 9:

(Study hours ≤ 9) $\Rightarrow 1, 2, 3, 4$ (2, 6, 4, 6, 8)
Pass = 2, fail = 2

$$\begin{aligned} \text{Gini} &= 1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2 \\ &= 1 - 0.25 - 0.25 = \boxed{0.5} \end{aligned}$$

(Study hours > 9) $\Rightarrow 5$ (10) $\Rightarrow$ pass = 1

$$\text{Gini} = 1 - (1)^2 = \boxed{0}$$

Step 3: Calculate weighted average Gini Impurity for each split

Split 3:

$\leq 3 \Rightarrow 1$, $> 3 \Rightarrow 4$

$$\left(\frac{1}{5}\right) \times 0 + \left(\frac{4}{5}\right) \times 0.375 = \boxed{0.3}$$

Split 5:

$\leq 5 \Rightarrow 2$, $> 5 \Rightarrow 3$

$$\left(\frac{2}{5}\right) \times 0 + \left(\frac{3}{5}\right) \times 0 = \boxed{0}$$

Split 7:

$\leq 7 \Rightarrow 3$, $> 7 \Rightarrow 2$

$$\left(\frac{3}{5}\right) \times 0.445 + \left(\frac{2}{5}\right) \times 0 = \boxed{0.266}$$

Split 9:

$\leq 9 \Rightarrow 4$, $> 9 \Rightarrow 1$

$$\left(\frac{4}{5}\right) \times 0.5 + \left(\frac{1}{5}\right) \times 0 = \boxed{0.4}$$

Step 4: Lowest weighted Gini is 0 at split point 5

